

Research paper

A robust model predictive control-based motion planning method for dual-arm I-AUVs cooperative operations

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ABSTRACT

This paper proposes a robust model predictive control (MPC) framework for cooperative motion planning of dual-arm intervention autonomous underwater vehicles. Addressing the challenges of cooperative optimization model for redundant system underwater operations, we introduce a hierarchical task decomposition strategy that classifies dual-arm cooperation into loose, partially constrained, and fully constrained modes. Each mode is further structured into tracking, coordination, and interaction subtasks, enabling a flexible representation of cooperative modes. To integrate these subtasks, a weighted subtask multi-objective MPC (WS-MOMPC) framework is developed in the work space, where task-specific weight matrices are assigned over six degrees of freedom to adapt to mission requirements. Moreover, to enhance robustness against ocean currents and model uncertainties, a dual-module tube MPC (DTMPC) strategy is proposed. The method maps tube constraints from state space to joint space and incorporates an auxiliary MPC-based compensation module to ensure precise trajectory tracking under disturbances. Simulation results validate that the proposed approach significantly improves motion stability, planning accuracy, and motion cost in complex underwater environments, thereby providing a scalable planning framework for future underwater dual-arm manipulation tasks.

1. Introduction

In recent years, underwater operational tasks are primarily conducted using Remotely Operated Vehicles (ROVs) equipped with robotic arms and manned submersibles (Petillot et al., 2019). However, researchers (Sivčev et al., 2018; Sun et al., 2025) have highlighted limitations in single-arm ROVs when executing complex underwater operations such as maintenance, docking, and sampling. In contrast, the use of intervention Autonomous Underwater Vehicles (I-AUVs) can greatly reduce operation time, operator risk, and cost (Nauert and Kampmann, 2023). Among these, dual-arm I-AUVs can overcome challenges like joint limitation, reduced dexterity, and task monotony inherent in conventional manipulators (Park and Lee, 2016). Due to the facilitation of complex cooperative operations, dual-arm I-AUVs demonstrate significant application value in field of Underwater Vehicle Manipulator Systems (UVMS) and emerge as a focal point in autonomous underwater operations research.

However, while dual-arm I-AUVs enhance operational flexibility, their redundant degrees of freedom introduce substantial challenges in motion planning, including configuration multiplicity, complex

cooperative optimization modeling, and significant disturbances from dynamic underwater environments. To address motion planning challenges, a wide range of robotic planning algorithms have been developed, which can be broadly categorized into four classes: graph-search-based, curve-interpolation-based, sampling-based, and optimization-based approaches (Yu et al., 2022b; Zhou et al., 2022).

Graph-search methods, such as Dijkstra and A*, plan by constructing grid maps to identify minimal-cost paths in static 2D or 3D environments. Enhanced variants, including hybrid A*, improved D* Lite, and anytime dynamic A* et al. (Dang et al., 2022; Maw et al., 2020; Yu et al., 2022a) have been developed to improve computational efficiency under dynamic conditions; however, their real-time performance remains limited in dynamic environments.

Curve-interpolation methods generate continuous trajectories using keyframes with Bezier, B-spline, or polynomial curves. Advanced techniques like multi-segment low-order Bezier approximation, triple NURBS interpolation, and quintic Bezier-polynomial hybridization et al. (Li et al., 2023; Ö and Tiemessen, 2022; Zhang et al., 2023) improve trajectory accuracy and computational efficiency while maintaining smoothness. Hybrid approaches integrating genetic algorithms (Yu

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et al., 2022c) or quadratic programming (Riboli et al., 2023) further optimize temporal and spatial path quality. However, these Curve-interpolation methods struggle to derive interpolation functions that satisfy the complex constraints of redundant systems.

Sampling-based methods such as PRM and RRT rapidly generate paths in high-dimensional spaces via random sampling and feasible tree construction. Due to the excellent adaptability to complex environment, sampling-based methods have been widely used in operations with manipulators. Improved methods like Cyl-HRRT*, Grounding-aware RRT* and EP-RRT* et al. (Ding et al., 2023; Enevoldsen and Galeazzi, 2021; Yu et al., 2023b) address optimality limitations but still fail to plan stable cooperative motion models for multi-manipulator systems. Optimization-based planning methods fundamentally aim to minimize or maximize an objective function under predefined constraints, thereby computing an optimal motion trajectory.

Among optimization-based approaches, MPC iteratively solves optimal actions at each time step based on predictive models, ensuring constraint satisfaction and optimization objectives. Its real-time cyclic planning enables adaptation to dynamic environments, making MPC widely adopted for complex constrained motion planning. For instance, autonomous lateral vehicle trajectory planning framed as an MPC problem to achieve stable parking in confined spaces (Kim et al., 2023); a human-like decision-making MPC framework proposed for anthropomorphic path planning (Hang et al., 2021); and a dual-arm welding robot system designed as coordinated motion objectives and constraints in MPC framework (He, 2019).

For underwater robot research, building on the citation (Whitney, 1969) which proposes the concept of solving optimal kinematic inverse solutions through constraint construction, the motion planning methods such as the minimum norm solution (Cho et al., 2017; Yanfu et al., 2008), gradient projection method (Jonghui and Wan Kyun, 2007; Li et al., 2019), and task priority method (Cieslak et al., 2015; Han and Chung, 2014; Yu et al., 2023a) have become the main approaches in UVMS coordinated motion planning. However, the dual-arm I-AUV motion planning faces three critical challenges: (1) complex optimization modeling for highly redundant cooperative systems; (2) multi-objective and multi-constraint planning framework of dual-arm I-AUV cooperative motion; (3) dynamic disturbances arising from underwater environment and internal system coupling. Traditional methods cannot holistically address the combined requirements of "high redundancy modeling + multi-constraint execution + robustness". Therefore, optimization-based MPC method, benefiting by its advantages of solving multi-constraint and multi-variable problems, is usually chosen to solve the motion planning of underwater operations.

In fact, as a robust control algorithm, MPC has been widely applied in underwater robot control systems for many years. Researchers develop various robust control algorithms by integrating deep learning (Cai et al., 2021), neural networks (Liu et al., 2025), and auxiliary control methods (Hao et al., 2024) within the MPC framework to meet task-specific operational requirements. In the field of underwater motion planning, MPC has been predominantly applied to high-speed AUVs requiring real-time obstacle avoidance (Yu et al., 2025), AUV swarms with multi-objective multi-constraint coordination (Zhao et al., 2024), and redundant UVMS for capture operations (Dai et al., 2019). Although prior studies have demonstrated the advantages of MPC in handling real-time processing, multi-constraint optimization, system redundancy, and disturbance robustness, they have yet to establish a unified framework for motion planning in redundant systems. Particularly, the dual-arm I-AUVs effectively formulating cooperative motion planning problems and deriving coordinated, robust solutions under multiple constraints and objectives remains an open challenge. To address these issues, this paper proposes a dual-module tube-based MPC (DTMPC) framework for cooperative motion planning of dual-arm I-AUVs. In this scheme, contributions are listed as:

- (a) To address the challenge of modeling complexity in dual-arm I-AUV cooperative motion planning, this study establishes a hierarchical planning framework that classifies cooperative motion patterns into two modes: a loose cooperative mode, which allows the dual arms to execute tasks with relative independence, and a tight cooperative mode, which enforces cooperative constraints between the arms. Each mode is further decomposed into tracking, coordination, and interaction subtasks, enabling flexible motion mode representation. By dynamically configuring these subtasks, the framework constructs a detailed optimization model that effectively reduces planning complexity while improving stability and adaptability in dynamic underwater environments.
- (b) A WS-MOMPC cooperative planning framework for dual-arm I-AUVs is developed, incorporating a multi-objective weighted combination model in the workspace. A motion cost function, derived from motion precision requirements and hydrodynamic resistance estimation, serves as a soft constraint to optimize vehicle-manipulator coordination. Hard constraints, including joint physical limitations, ensure operational safety in underwater local environment. The proposed WS-MOMPC framework satisfies both multi-objective optimization and stringent safety requirements for cooperative missions, as validated through simulation results.
- (c) An improved Tube MPC method is introduced to current disturbances and vehicle-manipulator coupling effects. By mapping "Tube" constraints from state-space to joint-space and designing an auxiliary MPC-based disturbance compensation solver module, the system achieves precise motion planning through redundant degrees of freedom utilization. Compared to conventional approaches, the DTMPC demonstrates well performance in redundant systems: joint-space Tube constraints enable coordinated disturbance rejection, while MPC optimization ensures strict adherence to motion limits, preventing overshoot and infeasibility. Simulation results show that DTMPC significantly improves disturbance rejection capability and operational stability during dual-arm cooperation.

The remainder of this paper is structured as follows. Section 2 presents the system modeling of the dual-arm I-AUV. Section 3 details the decomposition of cooperative motion modes and the formulation of subtasks. A novel DTPMC motion planning method will be proposed to realize robust underwater cooperative operations in Section 4. The simulation will be analyzed in Section 5 and we will make conclusion in Section 6.

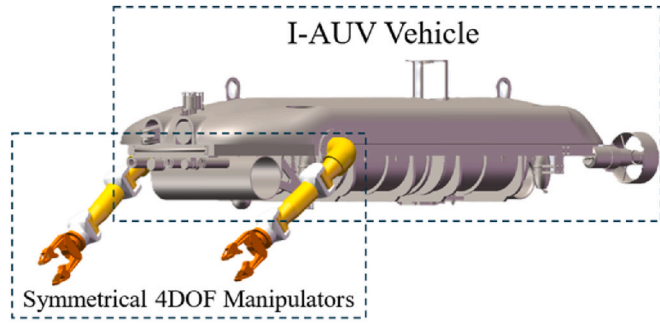
2. Dual-arm I-AUVs system model

The dual-arm I-AUV system studied in this paper is shown in Fig. 1 (a), consisting of an AUV vehicle and two isomorphic symmetric four-joint manipulators. As shown in Fig. 1(b) the robot obtains the relative position between the vehicle coordinate system $\{H\}$ and the object coordinate system $\{O\}$ through onboard sensors.

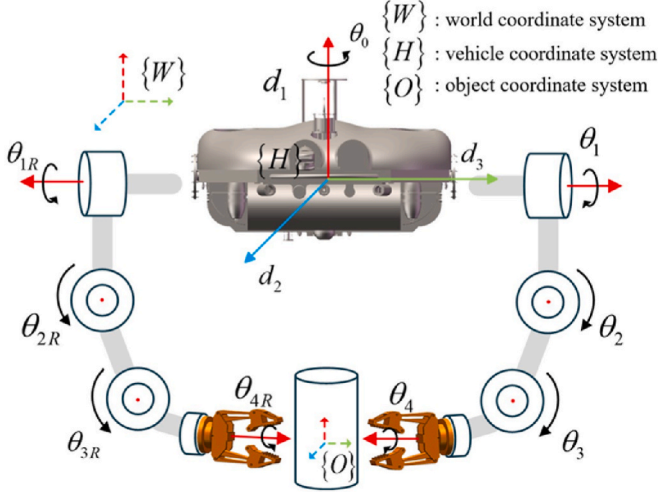
For the motion planning of the studied operational tasks, we consider that the roll and pitch degrees of freedom which may lead to unstable working states are excluded from the planning process. The AUV vehicle maintains a stable horizontal posture through control algorithms. Therefore, the dual-arm I-AUV system has:

- i) 4 degrees of freedom for the AUV vehicle: surge, sway, heave and yaw;
- ii) 4 degrees of freedom for each manipulator arm: shoulder rotation, two elbows rotation and wrist rotation.

In kinematic modeling, translational displacements (surge/sway/heave) are treated as prismatic joints, while rotational movements



(a) I-AUV vehicle and manipulators



(b) System coordinate system setting

Fig. 1. The model of dual-arm I-AUV.

(yaw/shoulder/elbow/wrist) are modeled as revolute joints. These joint motions are defined by $\theta = [d_1 \ d_2 \ d_3 \ \theta_0 \ \theta_1 \ \theta_2 \ \theta_3 \ \theta_4 \ \theta_{1R} \ \theta_{2R} \ \theta_{3R} \ \theta_{4R}]$.

For dual-arm I-AUVs, underwater operation motion planning involves adjusting the position and posture of the end-effectors at both arms' ends. Therefore, this study establishes an integrated kinematic model by combining the kinematic models of the vehicle body and the dual arms, linking the degrees of freedom motion of the dual-arm I-AUV system with the status of the arm ends. The posture of the robot's dual-arm end-effectors is solved as the state variable $\mathbf{x} = [\mathbf{R}_i \ \mathbf{Q}_i]$, $i = 1, 2$ through the joint motion variable θ of the left and right arms, where $\mathbf{R}_i$ represents the end positions of the left and right arms, and $\mathbf{Q}_i$ represents their end postures.

Based on the improved D-H method (Craig, 2009), the integrated kinematic model of the dual-arm I-AUV system is established in the APPENDIX and we can obtain:

$${}^0\mathbf{T}_n^i = {}^0\mathbf{T}_1^i \mathbf{T}_2^i \cdots \mathbf{T}_{n-1}^i \mathbf{T}_n^i = \begin{bmatrix} r_{11}^i & r_{12}^i & r_{13}^i & p_x^i \\ r_{21}^i & r_{22}^i & r_{23}^i & p_y^i \\ r_{31}^i & r_{32}^i & r_{33}^i & p_z^i \\ 0 & 0 & 0 & 1 \end{bmatrix}, i = 1, 2, \quad (1)$$

where: $p_x^i, p_y^i, p_z^i, p_x^i, p_y^i, p_z^i$ represents the position of dual-arm end-effectors.

Combining the system's kinematic model, the Z-Y-Z Euler angle calculation formula is as follows (Craig, 2009):

$$\begin{aligned} \beta &= \text{Atan } 2\left(\sqrt{(r_{31})^2 + (r_{32})^2}, r_{33}\right) \\ \alpha &= \text{Atan } 2(r_{23}/\sin \beta, r_{13}/\sin \beta) \\ \gamma &= \text{Atan } 2(r_{32}/\sin \beta, -r_{31}/\sin \beta) \end{aligned} \quad (2)$$

The state variables of the dual-arm I-AUV's left and right manipulators are denoted as:

$$\mathbf{x}^i(k) = [p_x^i \ p_y^i \ p_z^i \ \alpha^i \ \beta^i \ \gamma^i], i = 1, 2. \quad (3)$$

Based on the integrated kinematic model of the system, a predictive model is further established. The state variable $\mathbf{x}$ is denoted as a function of the joint variables θ of the robotic system:

$$\mathbf{x} = \mathbf{F}(\theta). \quad (4)$$

According to Equation (4), the discrete-time predictive model for the time step T is derived using the forward Euler method (Estep, 2002), as follows:

$$\mathbf{x}(k+1) = \mathbf{x}(k) + T \cdot \frac{\partial \mathbf{F}(\theta(k))}{\partial \theta(k)} \dot{\theta}(k). \quad (5)$$

The relationship between the future state variable $\mathbf{x}(k+1)$, the current state variable $\mathbf{x}(k)$, and the current control input $\dot{\theta}(k)$ is obtained. Let $\mathbf{B} = T \cdot \frac{\partial \mathbf{F}(\theta(k))}{\partial \theta(k)}$, and assuming that the joint variable θ remains at its initial value at each optimization time step, we can derive:

$$\mathbf{x}(k+1) = \mathbf{x}(k) + \mathbf{B}\mathbf{u}(k). \quad (6)$$

Based on these equations, a dual-arm end-effector state prediction model can be derived using state-space equations:

$$\mathbf{X}(k+1) = \mathbf{F}_a \mathbf{x}(k) + \mathbf{G}_a \mathbf{U}(k), \quad (7)$$

$$\text{where } \mathbf{X}(k+1) = \begin{bmatrix} \mathbf{x}(k+1|k) \\ \mathbf{x}(k+2|k) \\ \vdots \\ \mathbf{x}(k+N_p|k) \end{bmatrix}, \mathbf{U}(k) = \begin{bmatrix} \mathbf{u}(k) \\ \mathbf{u}(k+1) \\ \vdots \\ \mathbf{u}(k+N_c-1) \end{bmatrix}, \mathbf{F}_a = \begin{bmatrix} \mathbf{B} & 0 & 0 \\ \vdots & \ddots & \vdots \\ \mathbf{B} & \cdots & \mathbf{B} \\ \vdots & & \vdots \\ \mathbf{B} & \cdots & \sum_{i=0}^{N_p-N_c} \mathbf{B} \end{bmatrix}, \mathbf{G}_a = \begin{bmatrix} \mathbf{I} \\ \vdots \\ \mathbf{I}^{N_p} \end{bmatrix}.$$

3. Problem formulation

3.1. Division of Dual-Arm Cooperative Operation Modes

Different from single-arm UVMS, which operates with a single motion mode, dual-arm I-AUVs feature more complex cooperative operation modes. In this study, the cooperative operation process of dual-arm I-AUVs is divided into two categories: loose cooperative motion and tight cooperative motion. The latter can be further subdivided into partially constrained cooperative motion and fully constrained cooperative motion. Specifically:

- i) Loose Cooperative Motion: As illustrated in Fig. 2(a), this mode occurs during the initial stage of task execution, where the left and right arms share the same base and independently complete their own task. The cooperative motion in the workspace only needs to consider the impact of the vehicle's movement on each arm's independent tasks, coordinating the motion between the vehicle body and the mechanical arms during planning.
- ii) Partially Constrained Cooperative Motion: As shown in Fig. 2(b), this mode emphasizes coordination and constraints in specific

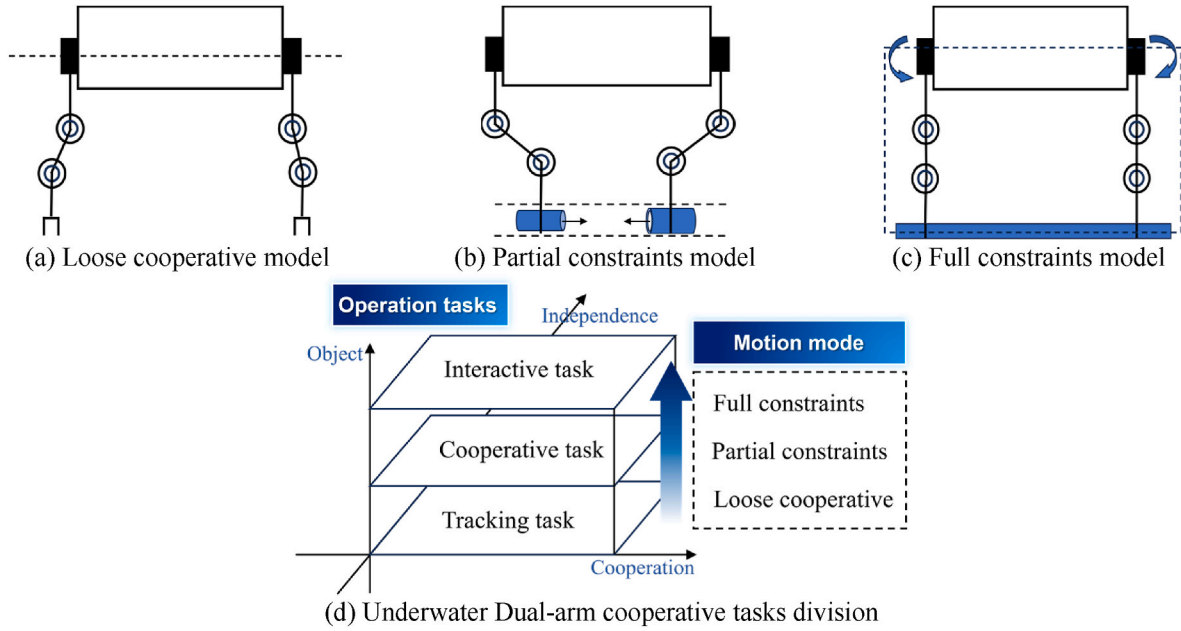


Fig. 2. Dual-Arm Cooperative Operation Modes and sub-tasks.

directions or tasks. The arms must synchronize movements in certain directions while remaining relatively independent in others.

- iii) Fully Constrained Cooperative Motion: As depicted in Fig. 2(c), this mode requires comprehensive consideration of constraints in all directions. The end effectors of both arms must strictly adhere to the cooperative task constraints in every direction, ensuring coordinated action with the object as a unified system.

These three operation modes cover almost all dual-arm cooperative operation scenarios of underwater dual-arm robots. However, due to the highly redundant degrees of freedom in the dual-arm I-AUV system, these three cooperative motion modes are difficult to express with a single task model. How to combine these three distinct types of subtasks to form various synergistic modes for dual-arm I-AUVs is the primary challenge addressed in this paper.

As shown in Fig. 2(d), this study decomposes the entire dual-arm cooperative operation motion planning task into a combination of three distinct motion subtasks, and achieves different motion modes by adjusting the proportion of each subtask. The definitions of the three subtasks are as follows:

Tracking Task: The end-effector states of the robot's dual arms are tracked and planned separately according to their respective target poses, with the objective of achieving the states specified by the target poses at the dual arms' end-effectors. The optimization cost function for the tracking task is defined as:

$$J_{trac}(k) = \mathbf{x}_r - \mathbf{x}(k), \quad (8)$$

where $\mathbf{x}_r$ represents the state variable of the target pose being tracked, $\mathbf{x}(k)$ represents the state variable of the robot system at the current moment.

Cooperative Task: The task involves two robotic arms considering each other's state variables within the workspace, cooperatively moving to achieve synergistic manipulation of a target object. The optimization cost function for the cooperative task is defined as:

$$J_{cop}(k) = \mathbf{x}_{left}(k) - \mathbf{x}_{right}(k), \quad (9)$$

where $\mathbf{x}_{left}(k)$ represents the state variables of the left arm's end-effector, $\mathbf{x}_{right}(k)$ represents the state variables of the right arm's end-effector.

Interactive Task: Considering the new system composed of arms-object, the interaction task is defined as the novel relationship between the end-effector states of the dual arms and the current state information of the object. The objective of this task is to achieve cooperative motion between the dual arms and the object. The optimization cost function for the interactive task, obeying Hooke's Law (Zhao et al., 2024), is defined as:

$$J_{obj}(k) = \mathbf{x}_{obj}(k) - \mathbf{x}(k), \quad (10)$$

where $\mathbf{x}_{obj}(k)$ represents the combined state of the end-effectors of the dual arms and the object when they form a new system.

The three cooperative motion modes together constitute a complete underwater dual-arm manipulation framework. Depending on the task scenario, the loose cooperative mode is employed when the two arms perform independent operations such as grasping separate targets, whereas the partially constrained or fully constrained motion modes are adopted when the arms need to cooperate closely on a shared task. Further, the mode transitions are determined based on the completion status of the preceding task phase and the evolution of the operational scene, forming a sequential execution strategy aligned with the overall mission objectives.

3.2. Underwater robust task

In practical underwater operations, as shown in Fig. 3, a wide range of disturbances—both from the environment and from internal system uncertainties—can impact the accuracy of robotic task execution.

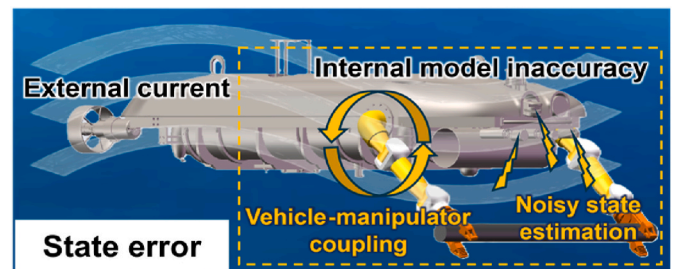


Fig. 3. State error caused by underwater disturbance.

Although these factors may act through the system's dynamics, their observable effect is a deviation in the system's kinematic output—specifically, the end-effector state error.

We therefore describe these errors as residual state deviations caused by internal and external sources:

External factors include hydrodynamic effects such as water currents, turbulence, and fluid interactions, which directly affect the kinematic state and cause the state error $\mathbf{e}_{out}^w$.

Internal factors originate from modeling errors, manipulator–vehicle coupling, and sensor inaccuracies, which also lead to kinematic state deviations and cause the state error $\mathbf{e}_{in}^w$. We define the system actual model $\mathbf{J}^w(\theta) = \mathbf{J}(\theta) + \mathbf{J}^{in}(\theta)$.

The dual-arm I-AUV kinematic model under such disturbances can be formally decoupled into a nominal state model and an additive state error term.:

$$\begin{aligned}\mathbf{x}(k+1) &= \mathbf{I}\mathbf{x}^*(k) + T \cdot \mathbf{J}^w(\theta) \dot{\theta}^*(k) + \mathbf{e}_{out}^w \\ &= \mathbf{I}\mathbf{x}^*(k) + T \cdot (\mathbf{J}(\theta) + \mathbf{J}^{in}(\theta)) \mathbf{u}^*(k) + \mathbf{e}_{out}^w \\ &= \mathbf{A}\mathbf{x}^*(k) + \mathbf{B}\mathbf{u}^*(k) + \mathbf{e}_{in}^w + \mathbf{e}_{out}^w \\ &= \mathbf{A}\mathbf{x}^*(k) + \mathbf{B}\mathbf{u}^*(k) + \mathbf{e}^w.\end{aligned}\quad (11)$$

Therefore, to compensate the state error $\mathbf{e}^w$ between the undisturbed nominal system state $\mathbf{x}^*(k)$ and the actual system state $\mathbf{x}(k)$ under disturbances, we define the system the robust motion planning task $\mathbf{J}_r$ as follows:

$$\mathbf{J}_r(k) = \mathbf{x}^*(k) - \mathbf{x}(k). \quad (12)$$

in contrast to conventional control strategies, which typically respond to disturbances through dynamic feedback and force-level control, we define the robust motion planning task in the kinematic-level residuals. We aim to enhance system robustness by optimizing the underwater robust task such that disturbance effects, once manifested as deviations in end-effector states, are compensated by motion adjustments at the planning level.

4. Improved TMPC-based planning method

In this section, we integrate the concept of tube MPC with the characteristics of dual-arm I-AUV coordinated motion to design an improved TMPC-based motion planning method tailored to the different task requirements discussed in section. The improved TMPC-based planning method partitions the planning process into two stages: First, a multi-task cooperative planning framework under ideal conditions is designed as a nominal system to satisfy cooperative operation modes and generate ideal actions. Second, a motion planning scheme for the disturbed system is developed based on robust task requirements to compensate for state errors arising from disturbances.

4.1. Nominal MOMPC-based cooperative planning

In the system nominal planning, the main objective is to realize the cooperative operation mode of dual-arm I-AUVs, ensuring that the robots can move according to the designed cooperative pattern in an ideal state. Inspired by the multi-objective optimization concept of WS-MOMPC (Shen et al., 2019), we construct a multi-tasks model to coordinate the three types of cooperative subtasks described in Section 3.1, thereby combining to achieve three types of cooperative motion modes. Specifically:

A. Cooperative task design:

In the cooperative tasks of dual-arm I-AUVs, the WS-MOMPC method can effectively coordinate three sub-tasks, thereby achieving coordinated motion planning for the robot's dual arms. Considering the tracking task, coordination task, and interaction task, the dual-arm

cooperative task model $\mathbf{J}_{multi}$ is designed as:

$$\mathbf{J}_{multi}(k) = \text{diag}(\mathbf{N}_{track}^i, \mathbf{N}_{cop}^i, \mathbf{N}_{obj}^i) \begin{bmatrix} \mathbf{J}_{trac}(k) \\ \mathbf{J}_{cop}(k) \\ \mathbf{J}_{obj}(k) \end{bmatrix}, \quad (13)$$

where the design of the weight matrixes is as follows:

$$\begin{aligned}\mathbf{N}_{track}^i &= \text{diag}(\eta_{tx}^i, \eta_{ty}^i, \eta_{tz}^i, \eta_{ta}^i, \eta_{tb}^i, \eta_{tr}^i), \eta_t^i \in [0, 1] \\ \mathbf{N}_{cop}^i &= \text{diag}(\eta_{cx}^i, \eta_{cy}^i, \eta_{cz}^i, \eta_{ca}^i, \eta_{cb}^i, \eta_{cr}^i), \eta_c^i \in [0, 1], \\ \mathbf{N}_{obj}^i &= \text{diag}(\eta_{ox}^i, \eta_{oy}^i, \eta_{oz}^i, \eta_{oa}^i, \eta_{ob}^i, \eta_{or}^i), \eta_o^i \in [0, 1]\end{aligned}\quad (14)$$

where η represents the sub-task weight coefficients corresponding to each state variable in the workspace.

Therefore, in the workspace, the optimization objectives for the dual arms in terms of the state variables are:

$$\mathbf{J}_d^i = \eta_{td}^i(d_r - d) + \eta_{cd}^i(d_{left} - d_{right}) + \eta_{od}^i(d_{obj} - d), \quad (15)$$

where $d = x, y, z, \alpha, \beta, \gamma$ represents the 6 degrees of freedom of the workspace.

formula (15) combines the end-effector states with the target states and the states between the dual arms in a weighted manner, using different weights to control the coordination of states across the 6-DOF workspace. The cooperative task model constructed based on WS-MOMPC can ensure end-effector tracking accuracy while maintaining coordinated movement between the dual arms, and incorporates new system requirements related to object interaction during the operation state.

B. Soft constraint based on motion cost optimization:

In addition to task model design for cooperative motion modes, how to allocate degrees of freedom between the manipulators and the vehicle, and coordinate their relationship, is a critical issue for underwater operations. As Fig. 4(a) shown, we generally define the ideal hand-vehicle coordination motion process as follows: when far from the target pose, the vehicle drives the motion, with the manipulator following and adjusting. With the target is approached, the manipulator's joint adjustments gradually take precedence for fine-tuning in the local vicinity. To achieve this, we define motion costs for different degrees of freedom in the planning phase, utilizing priority levels of internal degrees of freedom to tailor actions during various operational stages. As Fig. 4(b) shown, this includes two components: motion accuracy cost $\mathbf{M}$ and joint motion cost $\mathbf{F}$.

Motion accuracy cost $\mathbf{M}$ is defined as a weight matrix that systematically accounts for the differential impacts of the vehicle's translational motion and the manipulator's rotational movements on end-effector positioning precision. By increasing the weight assigned to vehicle displacement in the optimization process, an effective trade-off can be achieved between rapid vehicle locomotion and precise manipulator pose adjustment, thereby achieving the desired tracking performance for dual-arm I-AUV systems. The motion accuracy cost $\mathbf{M}$ is expressed as:

$$\mathbf{M} = \text{diag}(\mathbf{M}_{ve}, \mathbf{M}_m), \quad (16)$$

where $\mathbf{M}_{ve}$ represents the weight submatrix of the vehicle; $\mathbf{M}_m$ represents the weight submatrix of manipulators.

The joint motion cost $\mathbf{F} = \text{diag}(\mathbf{F}_{ve}, \mathbf{F}_m)$ is formulated based on hydrodynamic resistance estimation during robotic maneuvers, which establishes priority hierarchies for both the vehicle hull and manipulator joints. This strategy minimizes adjustments to joints with higher motion costs in planning, thereby reducing overall system energy expenditure.

For the vehicle, the approximate hydrodynamic resistance f_{ve} is derived from empirical formulae (Fossen, 2011):

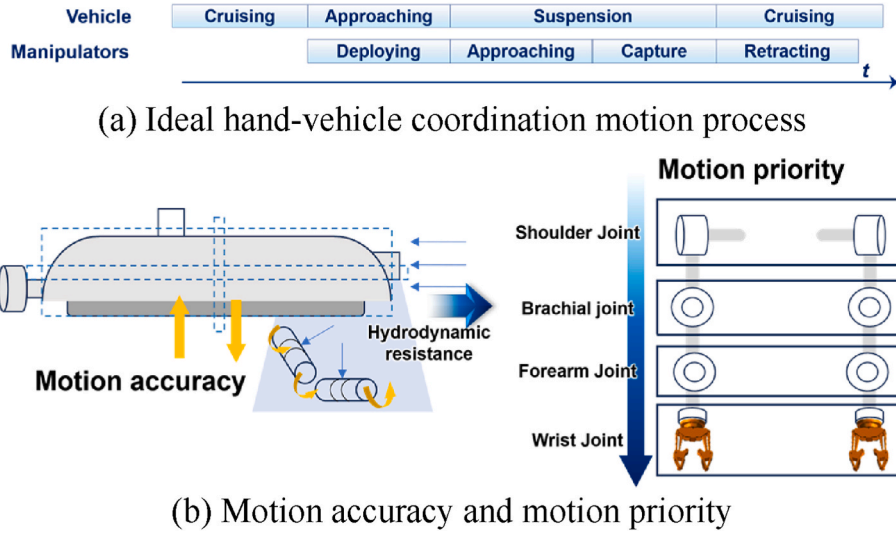


Fig. 4. The design of motion cost.

$$f_{ve} = \sum_{i=1}^3 \frac{1}{2} \rho C_{fi} A_i \dot{d}_i^2 = \sum_{i=1}^3 \frac{1}{2} f_{di} \Delta u_{di}(k)^2 \quad (17)$$

$$\approx \Delta \mathbf{u}_d(k)^T \text{diag}(f_{d1}, f_{d2}, f_{d3}) \Delta \mathbf{u}_d(k),$$

where $\dot{d}_i$ represents the three-degree-of-freedom moving speed of the vehicle; A represents the incoming area of the boat body; ρ stands for environmental water density; C_{fi} represents the water resistance coefficient in different directions of motion; $\Delta \mathbf{u}_{di}(k)$ represents the current vehicle's movement; f_{di} represent the motion cost weight corresponding to each direction movement.

The near-water resistance formula of the boat body is a formula about the moving speed of the vehicle in 3-DOF, and the water resistance generated during the displacement of the boat body in three directions is related to the oncoming area of the robot model in three directions and the water resistance coefficient. Therefore, we define the joint motion cost of the boat body in the planning:

$$\mathbf{F}_{ve} = \text{diag}(f_{d1}, f_{d2}, f_{d3}). \quad (18)$$

For manipulators, the water resistance torque due to wrist joint rotation is neglected by default, and thus is assigned the highest priority in the planning process. The approximate water resistance torques for each joint of dual robotic arms are calculated based on the Morison formula (Morison et al., 1950):

$$f_m = \sum_{i=1}^3 \frac{1}{2} \rho C_d D_i \int_0^l \|\mathbf{V}^i(x)\| ([x \ 0 \ 0] \times \mathbf{V}^i(x)) dx \quad (19)$$

$$\approx \Delta \mathbf{u}_\theta(k)^T \text{diag}(f_{\theta1} + f_{\theta2} + f_{\theta3}, \dots, f_{\theta1}, 1) \Delta \mathbf{u}_\theta(k),$$

where C_d represents resistance coefficient; D_i represents the diameter of each joint; $\mathbf{V}^i(x)$ represents the speed of each joint; $\Delta \mathbf{u}_\theta$ represents the variation in joint rotational speed

Therefore, based on the Morison hydrodynamic drag formula, the following joint motion cost for manipulators is defined:

$$\mathbf{F}_m = \text{diag}(f_{\theta1} + f_{\theta2} + f_{\theta3}, \dots, f_{\theta1}, 1, f_{\theta1R} + f_{\theta2R} + f_{\theta3R}, \dots, f_{\theta1R}, 1), \quad (20)$$

where $f_{\theta i}$ represents each joint motion cost weights for dual arms.

The joint motion costs $\mathbf{F}$ reveal the differential impact of each joint on the system, enabling the reduction of energy consumption and coupling effects between the robotic arms and the vehicle during motion.

C. Hard constraint based on physical constraint:

The position constraints on each joint of the robot are imposed by defining upper bounds $\theta_{\max}$ and lower bounds $\theta_{\min}$ through the constraint setup. Achieving systematic position constraints for the robotic system, which is mathematically formulated as:

$$\theta_{\min} \leq \theta \leq \theta_{\max}. \quad (21)$$

The constraints related to joint velocity variations are formulated as follows:

$$\frac{\theta_{\min} - \theta(k-1)}{t} - \dot{\theta}(k-1) \leq \Delta \dot{\theta}(k) \leq \frac{\theta_{\max} - \theta(k-1)}{t} - \dot{\theta}(k-1), \quad (22)$$

where $\theta(k-1)$ represents the joint positions at the previous time instance; $\dot{\theta}(k-1)$ represents the joint velocity at the previous time instance.

The constraint formulation concerning the planning input variables $\Delta \mathbf{U}$ is derived by considering the future time instances predicted by the model predictive planner:

$$\mathbf{Q}_{p\min} \leq \mathbf{H}_p \Delta \mathbf{U} \leq \mathbf{Q}_{p\max}, \quad (23)$$

$$\text{where } \mathbf{H}_p = \begin{bmatrix} \mathbf{I}_{12} & & \\ \vdots & \ddots & \\ \mathbf{I}_{12} & \dots & \mathbf{I}_{12} \end{bmatrix}_{12N_c \times 12},$$

$$\mathbf{Q}_{p\min} = \frac{1}{t} \begin{bmatrix} \theta_{\min} - \theta(k-1) - t\dot{\theta}(k-1) \\ \vdots \\ \theta_{\min} - \theta(k-1) - t\dot{\theta}(k-1) \end{bmatrix}_{12N_c \times 12},$$

$$\mathbf{Q}_{p\max} = \frac{1}{t} \begin{bmatrix} \theta_{\max} - \theta(k-1) - t\dot{\theta}(k-1) \\ \vdots \\ \theta_{\max} - \theta(k-1) - t\dot{\theta}(k-1) \end{bmatrix}_{12N_c \times 12}.$$

The velocity constraints on each joint of the robot are imposed by defining upper bounds $\dot{\theta}_{\max}$ and lower bounds $\dot{\theta}_{\min}$ through the constraint setup, which is formulated as:

$$\dot{\theta}_{\min} \leq \dot{\theta} \leq \dot{\theta}_{\max}. \quad (24)$$

The constraints related to the joint velocity variations are transformed into:

$$\dot{\theta}_{\min} - \dot{\theta}(k-1) \leq \Delta \dot{\theta}(k) \leq \dot{\theta}_{\max} - \dot{\theta}(k-1). \quad (25)$$

The constraint formulation concerning the planning input variables $\Delta \mathbf{U}$ is derived by considering the future time instances predicted by the model predictive planner:

$$\mathbf{Q}_{v\min} \leq \mathbf{H}_v \Delta \mathbf{U} \leq \mathbf{Q}_{v\max}, \quad (26)$$

$$\text{where } H_v = \begin{bmatrix} I_{12} & & \\ \vdots & \ddots & \\ I_{12} & \cdots & I_{12} \end{bmatrix}_{12N_c \times 12}, \quad Q_{v\min} = \begin{bmatrix} \dot{\theta}_{\min} - \dot{\theta}(k-1) \\ \vdots \\ \dot{\theta}_{\min} - \dot{\theta}(k-1) \end{bmatrix}_{12N_c \times 12},$$

$$Q_{v\max} = \begin{bmatrix} \dot{\theta}_{\max} - \dot{\theta}(k-1) \\ \vdots \\ \dot{\theta}_{\max} - \dot{\theta}(k-1) \end{bmatrix}_{12N_c \times 12}.$$

D. The structure of nominal WS-MOMPC

As shown in Fig. 5, the nominal WS-MOMPC framework contains multi-objects model from cooperative subtasks and constraints from both inter-system and environment. Therefore, the optimized cost function is formulated as follows:

$$\begin{aligned} \min J(k) &= \sum_{k=0}^{N_p-1} \|J_{\text{multi}}(k+i)^T Q_a J_{\text{multi}}(k+i)\| + \sum_{k=0}^{N_c-1} \|\Delta u(k+i)^T R_a \Delta u(k+i)\|, \\ \text{s.t. } X(k+1) &= F_w x(k) + G_w U(k), A_{\text{cons}} \Delta U(k) \leq b_{\text{cons}}, |J(\theta)| \neq 0 \end{aligned} \quad (27)$$

where Q_a denotes the weight of the cooperative mission term, $R_a = MF$ corresponds to the motion cost optimization term incorporating joint-space motion priority design, $A_{\text{cons}} \Delta U(k) \leq b_{\text{cons}}$ signifies the physical constraint design for joints. The system Jacobian matrix $|J(\theta)| \neq 0$ is incorporated to avoid singular configurations.

4.2. Model-based ancillary MPC planning design

The traditional rigid tube-based TMPC (Rigid Tube MPC, RTMPC) method ensures system stability and safety under various uncertainty conditions by designing an accurate minimum robust positively invariant set (mRPI) (Rakovic et al., 2005). The mRPI is defined as the minimal set that guarantees system state containment within it under all possible disturbances, thereby providing robustness assurance for system design. Generally, the mRPI in rigid tube-based MPC methods is approximately computed using the following approach (Langson et al., 2004):

Consider a disturbed system model $x^* = Ax + w$, $w \in W$, where matrix A exhibits eigenvalues with magnitude less than 1. The mRPI for this system is mathematically formulated as:

$$F_\infty = \bigoplus_{i=0}^{\infty} A^i W, \quad (28)$$

where F_∞ denotes a set such that if the system state resides within F_∞ at a given moment, then it will remain in the set indefinitely for all subsequent time steps, irrespective of disturbances $w \in W$; $\oplus$ represents the

Minkowski sum; A^i corresponds to the system state after an infinite number of time steps.

Therefore, through iterative refinement processes, the approximate finite-time horizon mRPI is obtained:

$$F_s \triangleq \bigoplus_{i=0}^{s-1} A^i W, F_0 \triangleq \{0\}, \quad (29)$$

where F_s denotes the outer approximation of the mRPI set after a finite number of steps, serving as an estimate for F_∞ ; The symbol $\triangleq$ denotes "is defined as"; F_0 represents the set containing only the origin, characterizing the system state under disturbance-free conditions.

Furthermore, the external approximation of mRPI is obtained to ensure system state stationarity under disturbances. Employing $(1-\alpha)^{-1}$ scaling which is a conservative approach to guarantee system state containment within boundary constraints during perturbations:

$$F(\alpha, s) \triangleq (1-\alpha)^{-1} F_s, \quad (30)$$

where α represents a scaling factor less than 1 (typically greater than 0) that determines the conservatism of the approximation. A smaller scaling factor results in larger, more conservative sets.

The approximation of mRPI generally requires numerous iterations to ensure approximation accuracy. Therefore, in RTMPC methods, mRPI is typically offline-computed, limiting its capability as a controller to handle minor disturbances without dynamic programming performance. Moreover, computing accurate mRPI becomes computationally challenging for systems with numerous state variables. For underwater dual-arm operation motion planning, traditional RTMPC methods exhibit excessive conservatism, hard to satisfy the motion planning requirements for the redundant dual-arm I-AUV. Therefore, motion planning of UVMS usually choose TMPC methods combining with other auxiliary solvers, such as TMPC based on state feedback (FBTMPC) (Mayne and Langson, 2001).

For the underwater robust task proposed in Section 3.2, inspired by Tube-based Robust Nonlinear MPC framework (Mayne et al., 2011), we select a model-based MPC planner as an auxiliary solver to address internal and external disturbances generated during underwater operations. We have:

$$\begin{aligned} \min J_{\text{anci}}(k) &= \sum_{k=0}^{N_p-1} \|J_r(k+i)^T Q_{\text{anci}} J_r(k+i)\| + \|\Delta U(k)\|^2 + \underbrace{V_f(e)}_{\text{Terminal cost}}, \\ \text{s.t. } X(k+1) &= F_w x(k) + G_w U(k) \\ A_{\text{cons}} U(k) &\leq b_{\text{cons}} \end{aligned} \quad (31)$$

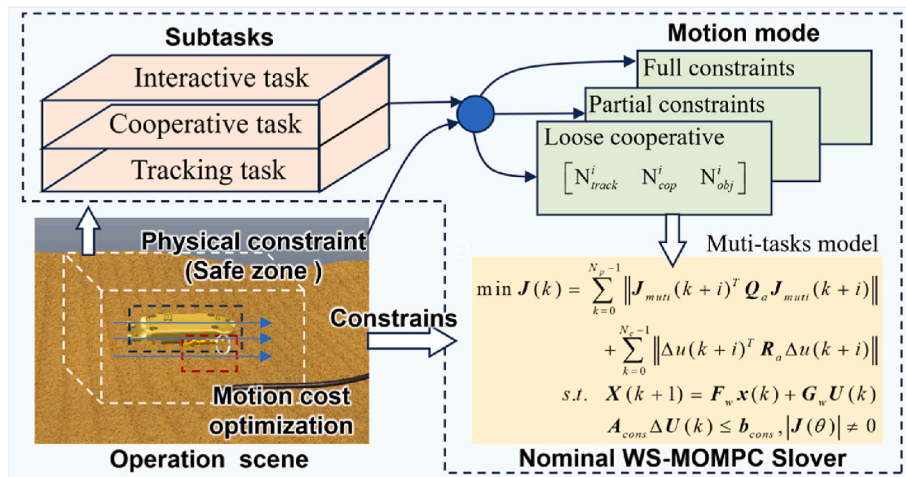


Fig. 5. Nominal WS-MOMPC framework.

where Q_{anci} denotes the weight of the robust mission term; the terminal cost $V_f(e) = e(k + N_p)^T P e(k + N_p)$.

Additionally, unlike single-arm UVMS requiring only manipulator-vehicle coordination, dual-arm I-AUV motion planning necessitates distinguishing joint-level motion priorities during cooperative movements to ensure redundant systems can generate disturbance-adaptive actions stabilizing the system. Therefore, we map the ‘Tube’ from state space to joint space.

Building upon conventional tube construction methods in generalized state spaces, we supplementally define Tube soft constraints in the joint space, where each joint’s motion is subject to corresponding tube constraints defined by cooperative motion patterns. This enables robust following of ideal trajectories while avoiding additional destabilizing disturbances. For dual-arm I-AUV dynamic capture missions, small-scale motion compensations induced by disturbances are deemed beneficial for system stability when executed through the vehicle’s movements.

$$\begin{aligned} \min \quad & J_{anci}(k) = \sum_{k=0}^{N_p-1} \|J_r(k+i)^T Q_{anci} J_r(k+i)\| + \sum_{k=0}^{N_c-1} \|\Delta u(k+i)^T W_{tube} \Delta u(k+i)\| + \underbrace{V_f(e)}_{\text{Terminal cost}} \\ \text{s.t.} \quad & X(k+1) = F_w x(k) + G_w U(k) \\ & A_{cons} U(k) \leq b_{cons} \end{aligned} \quad (33)$$

Therefore, vehicle motion prioritization exceeds joint priorities in the auxiliary solver design, preferentially adjusting vehicle motion to counteract disturbances.

The joint-space tube constraint allocations are specifically illustrated in Fig. 6. We further decompose the tube constraints in the state space and establish three types of motion priority-based soft constraints in the joint space corresponding to three motion modes: in loose coordination mode, the vehicle has higher priority than the manipulator; in partially constrained mode, free joints have higher priority than constrained joints; and in full constrained mode, constrained joints collectively maintain low priority. Overall, the tube design in the joint space primarily prioritizes hull compensation motion, while the manipulator maintains corresponding adjustments under constraints to preserve the stability of the overall system. We define corresponding motion priority tubes that adapt to changes in different cooperative motion patterns:

$$W_{tube}(\Omega(k)) = \begin{cases} W_l = \text{diag}(w_h, w_h, w_h, w_h, w_m, w_m, w_m, w_m), \Omega(k) = l \\ W_p = \text{diag}(w_h, w_h, w_h, w_h, w_l, w_m, w_m, w_l), \Omega(k) = p \\ W_f = \text{diag}(w_h, w_h, w_h, w_h, w_m, w_l, w_l, w_l), \Omega(k) = f \end{cases} \quad (32)$$

where W_l applicable to loose cooperative motion is typically assigned smaller values to prioritize vehicle-based disturbance rejection while minimizing coupling disturbances caused by repeated arm adjustments; W_p , designed for partially constrained cooperative modes, further constrains joint rotations of manipulators to prevent inefficient planning due to recurrent joint movements; W_f , designed for fully constrained cooperative modes, retains only shoulder joint rotations to maintain stable relative configurations between dual arms and objects; w represents three different levels of priorities: high, medium and low.

Therefore, the MPC-based auxiliary solver design primarily comprises two components: compensation motion solution terms for state errors between the nominal system and the disturbed system, and optimization terms for coordinated planning. The auxiliary solver design is formulated as follow:

4.3. The structure of DTMPC motion planning method

In summary, for the robust motion planning problem of the linear time-invariant dual-arm I-AUV model studied in this work, the DTMPC method can provide sufficient robustness. The proposed DTMPC-based motion planning framework is illustrated in Fig. 7.

The process of the proposed motion planning algorithm is shown as Fig. 8. At the beginning, the dual-arm I-AUV system selects the corresponding motion mode based on the scene task information. Then, obtain the states of underwater targets as the real-time planning input. In the nominal system, joint nominal inputs u^* satisfying both tracking-capture tasks and constrained cooperative requirements are obtained through the nominal solver based on target positions and robot’s real-time state quantities. The nominal system state x^* is derived from the ideal model of the nominal system. Then, by computing the difference between the actual system state x under disturbances and the nominal system state x^* , we can get the state error e :

$$e = x(k) - x^*(k). \quad (34)$$

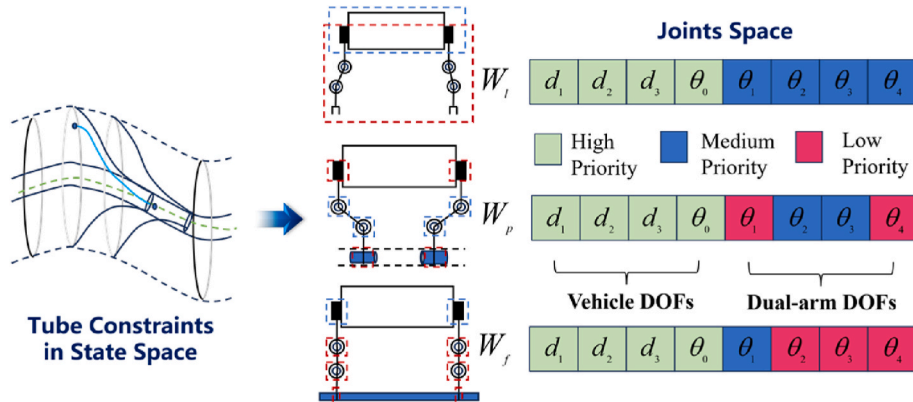


Fig. 6. The constraints allocation of joints space.

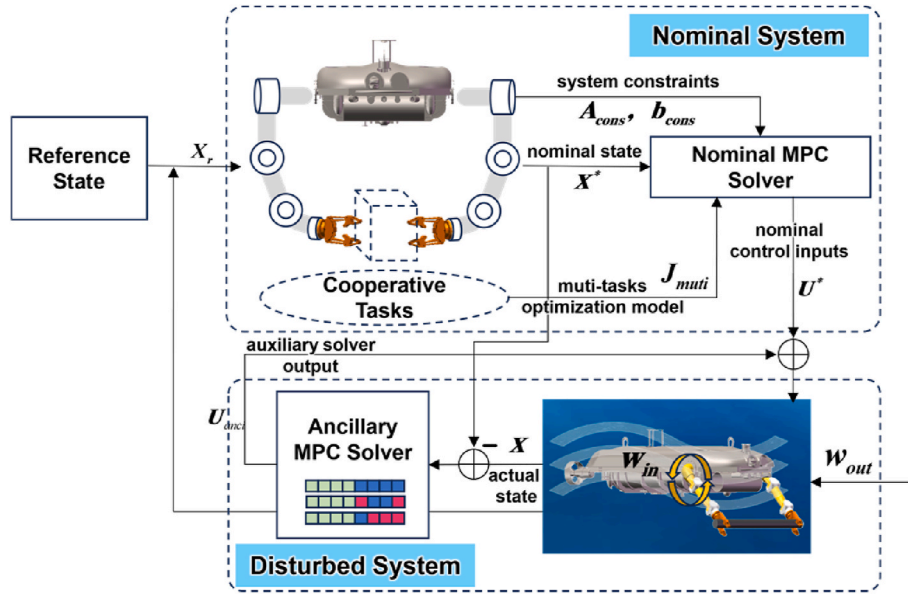


Fig. 7. DTMPC motion planning framework.

Subsequently, in the disturbed system, the obtained state error e is input into the MPC-based auxiliary solver to compute the auxiliary planning output u_{anci} . Finally, the planning inputs u^* derived from the nominal system and u_{anci} generated by the auxiliary solver jointly act on the actual system

$$u = u^* + u_{anci}. \quad (35)$$

At this stage, the actual disturbed system generates the current actual system state X , which, together with the real-time updated target state is fed back into the nominal system's planning solver to initiate the next planning cycle.

Algorithm 1. Improved Dual-module tube MPC

```

Initialize predictive datasets  $\{N_p, N_c\}$ 
Initialize cooperative motion mode weigh value  $\{N_{trac}, N_{cop}, N_{obj}\}$ 
for time = 0, 1 ..., t
    Update the current target state  $x_r$  and system state  $x$ 
    Calculate the optimal planning input  $u^*$  and nominal state  $x^*$  for the nominal
    system with (27)
    Update the state error  $e$  between the nominal system and the disturbed system
    with (34)
    Compute the compensatory planning input  $u_{anci}$  in auxiliary solver with (33)
    Obtain the actual planning action  $u$  applied to the system with (35)
end for
  
```

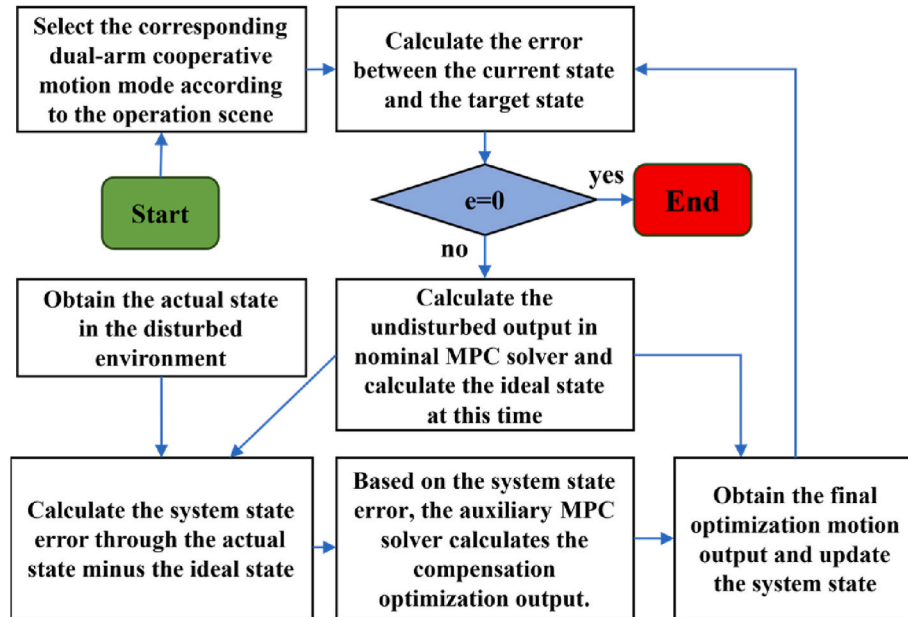


Fig. 8. The process of DTMPC motion planning algorithm.

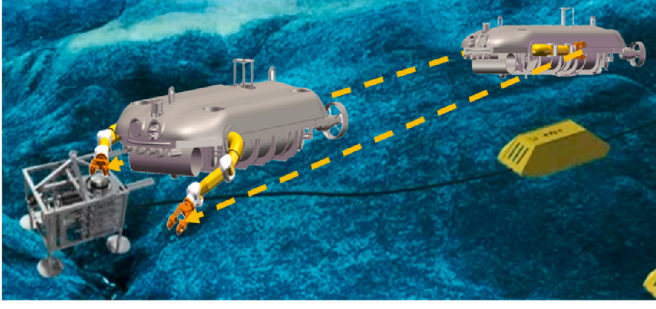


Fig. 9. Underwater facility maintenance.

4.3.1. DTMPC robustness analysis

Consider the discrete-time auxiliary MPC system:

$$\mathbf{e}(k+1) = \mathbf{A}\mathbf{e}(k) + \mathbf{B}\Delta\mathbf{u}^*(k) + \mathbf{B}\mathbf{u}_v(k) + \mathbf{w}(k), \quad (36)$$

where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the state transition matrix; $\Delta\mathbf{u}^*(k)$ denotes the output under ideal environment; $\mathbf{u}_v(k)$ denotes the deviation caused by bounded noisy measurement $\mathbf{v}(k)$, $\|\mathbf{v}(k)\| \leq \bar{\mathbf{v}}(k)$ $\mathbf{w}(k)$ is a bounded external disturbance satisfying $\|\mathbf{w}(k)\| \leq \bar{\mathbf{w}}(k)$.

Define a discrete-time Lyapunov function:

$$\mathbf{V}(\mathbf{e}(k)) = \mathbf{e}(k)^T \mathbf{P} \mathbf{e}(k). \quad (37)$$

We obtain:

$$\begin{aligned} \Delta \mathbf{V} &= \mathbf{V}(\mathbf{e}(k+1)) - \mathbf{V}(\mathbf{e}(k)) \\ &= \mathbf{e}(k)^T (\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P}) \mathbf{e}(k) + 2\mathbf{e}(k)^T \mathbf{A}^T \mathbf{P} (\mathbf{B}\delta\mathbf{u}^* + \mathbf{B}\mathbf{u}_v + \mathbf{w}) \\ &\quad + (\mathbf{B}\delta\mathbf{u}^* + \mathbf{B}\mathbf{u}_v + \mathbf{w})^T \mathbf{P} (\mathbf{B}\delta\mathbf{u}^* + \mathbf{B}\mathbf{u}_v + \mathbf{w}). \end{aligned} \quad (38)$$

For $\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P} \prec 0$, some $\alpha > 0$, the first term satisfies:

$$\mathbf{e}(k)^T (\mathbf{A}^T \mathbf{P} \mathbf{A} - \mathbf{P}) \mathbf{e}(k) \leq -\alpha \|\mathbf{e}(k)\|^2. \quad (39)$$

All other terms are either linear or quadratic in $\mathbf{u}_v(k)$ and $\mathbf{w}(k)$, which are bounded by $\bar{\mathbf{v}}(k)$ and $\bar{\mathbf{w}}(k)$. Therefore, we upper bound all remaining contributions as:

$$\Delta \mathbf{V} \leq -\alpha \|\mathbf{e}(k)\|^2 + \beta, \quad (40)$$

where β depending on the $\bar{\mathbf{v}}(k)$ and $\bar{\mathbf{w}}(k)$.

Therefore, $\mathbf{e}(k)$ converges exponentially to zero under bounded $\mathbf{v}(k)$ and $\mathbf{w}(k)$ which ensures convergence of the auxiliary model-based MPC solver and the overall system maintains controllable accuracy under bounded uncertainties. This proves the robustness of the full DTMPC system.

5. Simulation and preliminary experimental validation

5.1. The feasibility of the proposed cooperative subtasks planning

A. The feasibility of the loose cooperative motion planning

We choose underwater facility maintenance operations as the representation of loose cooperative motion planning. In this operation simulation, as Fig. 9 shown, each arm performs its own tasks and reaches their own targets. The parameters setting is shown as Table 1:

The simulation results demonstrate the effectiveness of the proposed manipulator-vehicle coordinated motion planning method for underwater dual-arm robots in executing loose coordination tasks. As shown in Fig. 10(a), the end-effector trajectory reveals insufficient coordination between the robotic arms and the vehicle when joint motion cost optimization is neglected, exhibiting disordered adjustments that lead to high motion costs and redundant actions. In contrast, the proposed method effectively coordinates task execution by optimizing joint velocity profiles, eliminating transient high-speed movements in the

planned trajectory, thereby significantly enhancing operational stability and trajectory precision. Fig. 10(b) illustrates the proposed method ensures that the system's motion cost decrease monotonically during loose cooperative motion planning.

The end-effector poses displayed in Fig. 10(c)–(f) demonstrate rapid convergence to target positions while achieving precise pose alignment through adaptive adjustments during approach phases.

Analysis of Fig. 10(g)–(j) reveals a two-stage motion planning process: an initial phase of rapid target approaching with synchronized pose correction, followed by a precise pose adjustment phase. This progression aligns with optimal strategies for underwater operations. Notably, the planning process exhibits no oscillatory position adjustments or local optimization traps, validating the effectiveness of the designed motion cost function and joint physical constraints. This ensures efficient arm-vehicle coordination, achieving notable improvements in both trajectory precision and computational efficiency.

B. The feasibility of the partially constrained cooperative motion planning

We choose underwater plug-pull operation as the representation of partially constrained cooperative motion planning. In this operation simulation, as Fig. 11 shown, the dual arms move to each other in same pose. The parameters setting is shown as Table 2:

Simulation results demonstrate that during the initial motion planning phase, the end-effector exhibits brief micro-displacements in the x and z -directions, which are rapidly stabilized through real-time adjustments while maintaining synchronization. The terminal pose also experiences initial minor oscillations but promptly converges to the target configuration and remains stably aligned, satisfying the requirements of partially constrained cooperative tasks. As shown in Fig. 12(a), after initial brief adjustments, both manipulators strictly follow the planned motion trajectory, maintaining positional synchronization in the x - and z -directions while independently moving in the y -direction to achieve designated docking positions. Fig. 12(b) and (c) demonstrates that as the y -direction spacing between end-effectors reduces, their terminal velocities gradually decrease, reflecting the impact of joint motion priority design on optimization outcomes and ensuring smooth deceleration during operation. Fig. 12(d) presents the joint-space position variations. Through joint motion priority settings, the vessel's pitch motion, y -direction displacement, and manipulator shoulder and wrist joint movements are significantly restricted, remaining nearly stationary. Meanwhile, the vessel's x and z -direction displacements coordinate with the manipulator's elbow and forearm joint motions to maintain stable end-effector postures and sustained relative motion. Fig. 12(e) illustrates the proposed method ensures that the system's motion cost

Table 1

Loose cooperative motion parameters.

Parameters	Simulation setting
predictive datasets	$\{N_p = 5, N_c = 1\}$
initial state	$[-1.05, 0.5, 0, 0.5, -0.5, -0.5, 0.5, -1.05, -0.5, 0, 0.5, -0.5, -0.5, 0.5]$
target state	$[1.86, -0.92, -2.72, 0.66, -0.31, -0.79, -0.44, -0.72, 2.87, -2.67, -2.72, 0.25, 0.53, 0.66, -0.05]$
tracking term weight matrix	$\mathbf{Q}_d = \text{diag}(1.2, 1.2, 1.2, 0.5, 0.5, 0.5, 0.5, 1, 1, 1, 0.5, 0.5, 0.5, 0.5)$
control variable weight matrix	$\mathbf{R}_d = \text{diag}(1.5 \times 10^3, 10^3, 10^3, 10^2, 10, 5, 2, 1, 10, 5, 2, 1)$
position constraint (upper limit)	$[2.5, 2.5, 2.5, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi]$
position constraint (lower limit)	$[-2.5, -2.5, -2.5, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi]$
velocity constraint (upper limit)	$[1, 1, 1, \pi/6, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18, \pi/18]$
velocity constraint (lower limit)	$[-1, -1, -1, -\pi/6, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18, -\pi/18]$

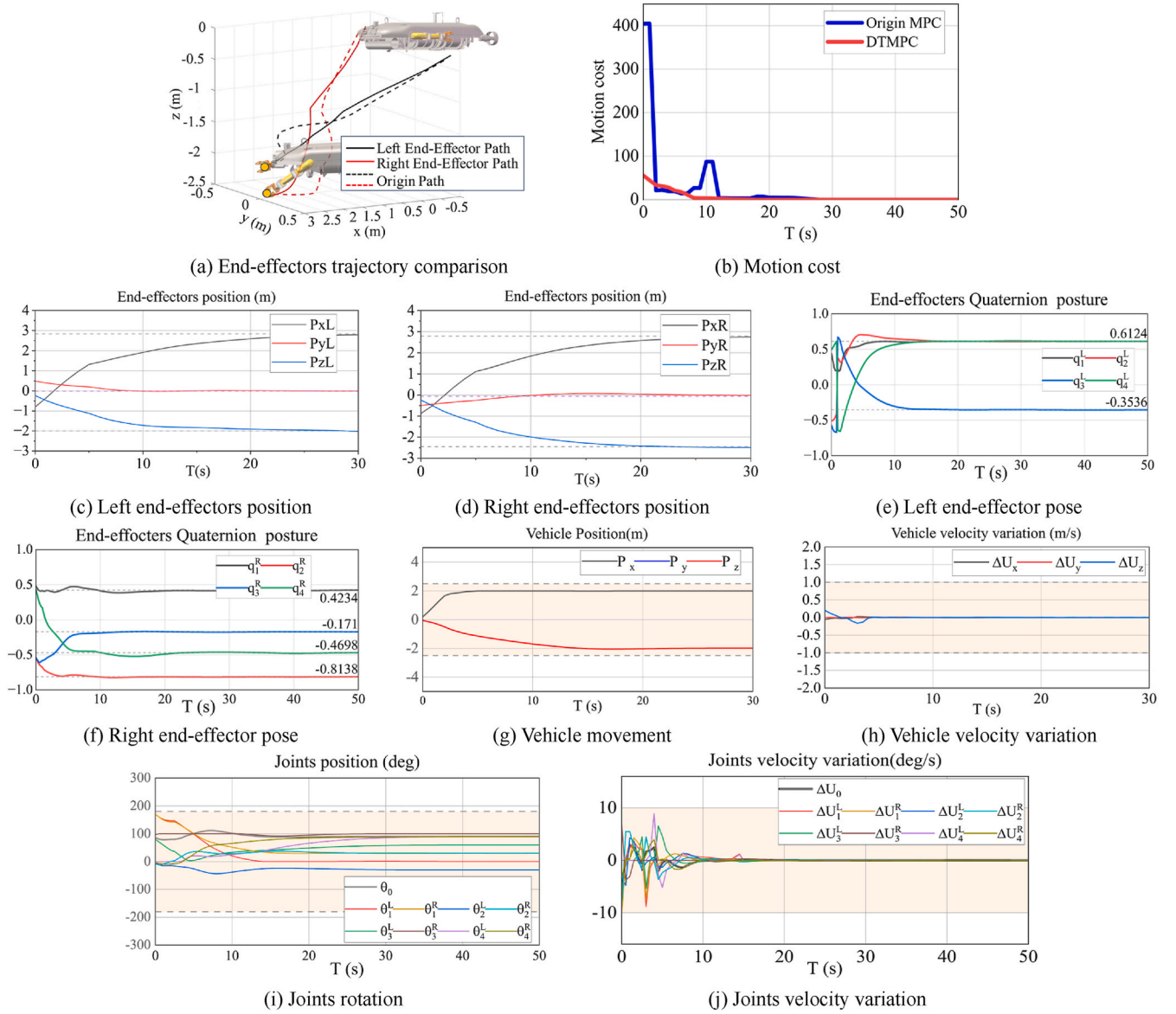


Fig. 10. Simulation results of loose cooperative motion.

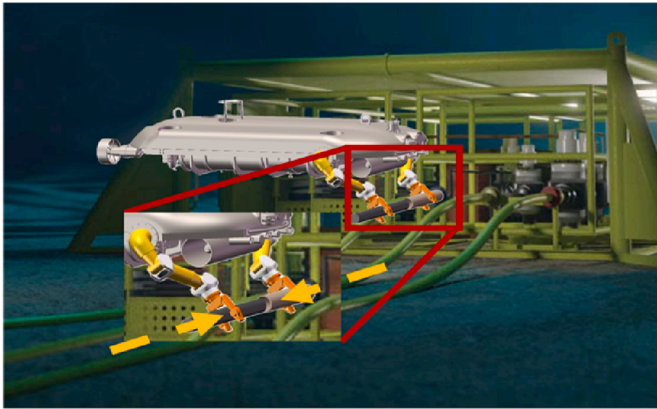


Fig. 11. Underwater plug-pull operation.

Table 2

Partially constrained cooperative motion parameters.

Parameters	Simulation setting
predictive datasets	$\{N_p = 5, N_c = 1\}; \mathbf{N}_{track}^i = \mathbf{I}_{12}; \mathbf{N}_{cop}^i = 0.5\mathbf{I}_{12}$
initial state	$[0.49, 0.75, -0.85, 2.62, 0, 1.57, 0.49, -0.75, -0.85, 2.62, 0, 1.57]$
target state	$[0.49, 0, -0.85, 2.62, 0, 1.57, 0.49, 0, -0.85, 2.62, 0, 1.57]$
weight of cooperative mission	$\mathbf{Q}_a = \mathbf{I}_{12}$
control variable weight matrix	$\mathbf{R}_a = \text{diag}(1.5 \times 10^3, 10^3, 10^3, 10^2, 10, 5, 2, 1, 10, 5, 2, 1)$
position constraint (upper limit)	$[2, 2, 2, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi]$
position constraint (lower limit)	$[-2, -2, -2, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi]$
velocity constraint (upper limit)	$[0.1, 0.1, 0.1, \pi/12, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36]$
velocity constraint (lower limit)	$[-0.1, -0.1, -0.1, -\pi/12, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36]$

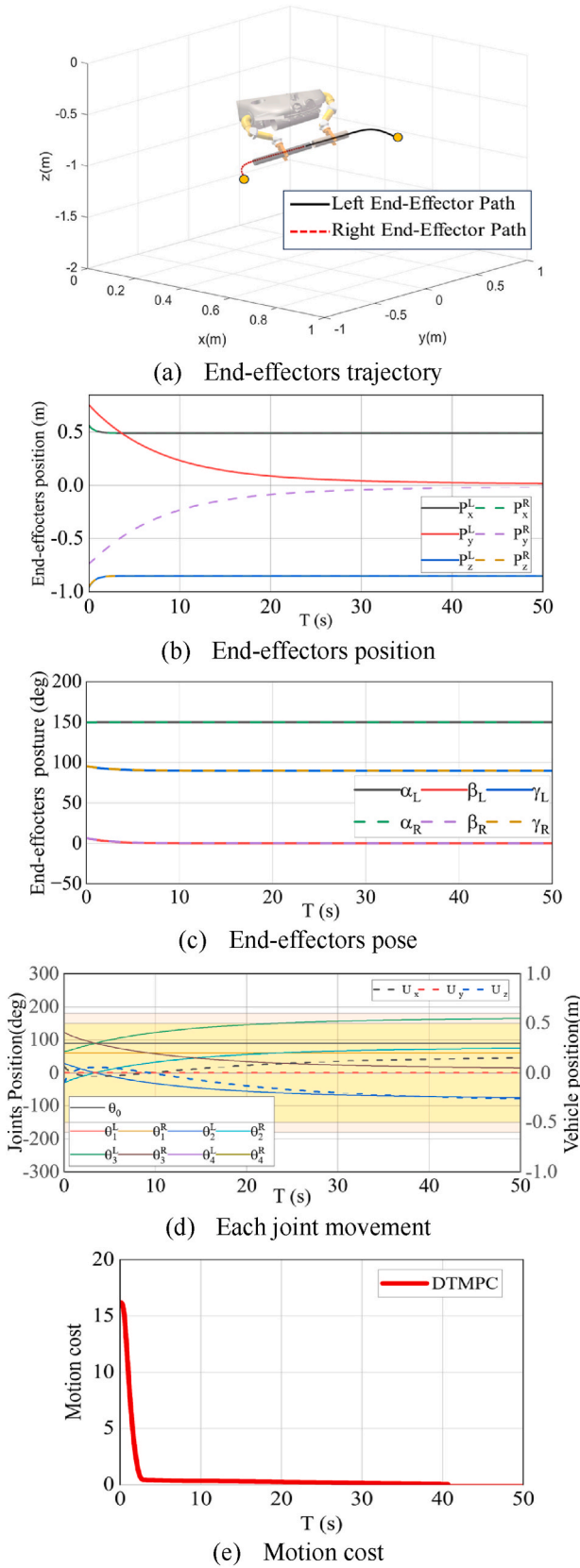


Fig. 12. Partially constrained cooperative motion simulation results.

C. The feasibility of the fully constrained cooperative motion planning

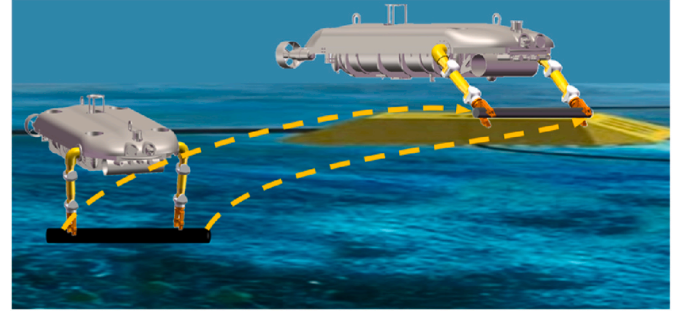


Fig. 13. Underwater target handling.

Table 3

Fully constrained cooperative motion parameters.

Parameters	Simulation setting
predictive datasets	$\{N_p = 5, N_c = 1\}$; $N_{track}^i = I_{12}$; $N_{cop}^i = 0.5I_{12}$; $N_{obj}^i = 0.5I_{12}$
initial state	$[0.16, 0.50, -1.04, 2.98, 0, 1.57, 0.16, -0.5, -1.03, 2.98, 0, 1.5708]$
target state	$[2.74, 0.5, 1.26, 2.36, 0, 1.57, 2.74, -0.5, 1.26, 2.36, 0, 1.57]$
tracking term weight matrix	$Q_a = I_{12}$
control variable weight matrix	$R_a = \text{diag}(1.5 \times 10^3, 10^3, 10^3, 10^2, 10, 5, 2, 1, 10, 5, 2, 1)$
position constraint (upper limit)	$[5, 5, 5, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi]$
position constraint (lower limit)	$[-5, -5, -5, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi]$
velocity constraint (upper limit)	$[1, 1, 1, \pi/6, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36]$
velocity constraint (lower limit)	$[-1, -1, -1, -\pi/6, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36]$

decrease monotonically during partially constrained cooperative motion planning.

We choose underwater target handling operation as the representation of fully constrained cooperative motion planning. In this operation simulation, as Fig. 13 shown, the dual arms move synchronously to carry the heavy object to target position. The parameters setting is shown as Table 3:

Simulation results demonstrate that during the initial stage of synchronized transportation tasks, the end-effectors rapidly and precisely reach the designated positions in the x and z -directions while maintaining positional synchronization. In the y -direction, the end-effector positions remain stable near the prescribed locations, exhibiting minor oscillations that do not compromise the coordinated clamping state of the dual arms, thereby avoiding interference with the object's internal structure. Fig. 14(a) shows that after initial adjustments, the end-effector trajectories strictly follow the planned motion, preserving synchronization in the x and z -directions while maintaining a constant inter-arm distance in the y -direction, successfully accomplishing the synchronized transportation task.

Fig. 14(b) and (c) reveal that the end-effector postures change synchronously, consistently fulfilling the cooperative requirements of fully constrained tasks. Fig. 14(d) presents the joint-space position variations, demonstrating that during planning, the vessel's pitch motion, y -direction displacement, and the manipulator's shoulder, elbow, and wrist joints are assigned low motion priorities, effectively suppressing their interference with the task. The simulation results show that these joints exhibit significantly restricted motion amplitudes, remaining nearly stationary to ensure task stability and successfully complete the synchronized transportation operation. Fig. 14(e) illustrates the proposed method ensures that the system's motion cost decrease monotonically during fully constrained cooperative motion planning.

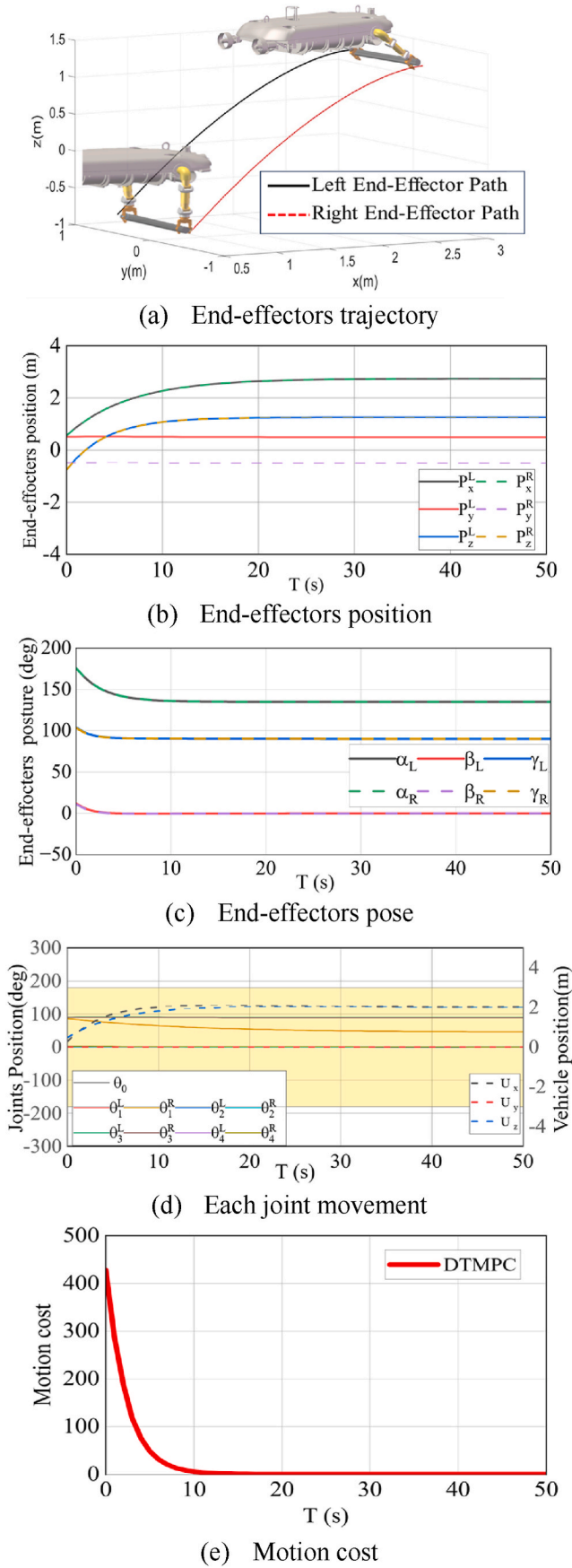


Fig. 14. Fully constrained cooperative motion simulation results.

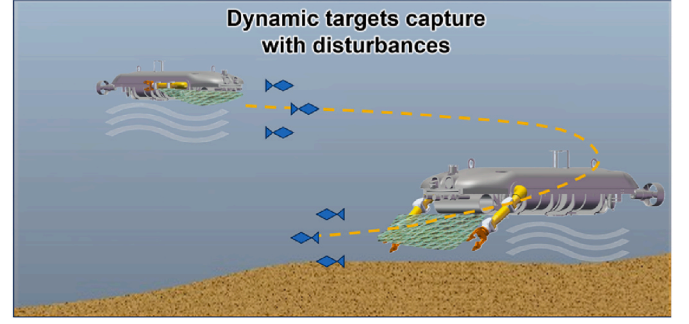


Fig. 15. Dual-arm capture operation.

5.2. Robust planning performance under disturbances

In the simulation design presented in this section, the task objective is to achieve underwater dynamic capture operations, as Fig. 15 illustrated. During mission execution, the robot must maintain a fully constrained motion mode while deploying dual-arm clamping fishing gear to accomplish capture of the dynamic target under hydrodynamic disturbances and system-coupled interference.

For the dynamic target, its motion state information is defined as follows:

$$\begin{cases} x_r = 5 \sin(0.3t) \\ y_r = 5 \cos(0.3t) \\ z_r = -0.5t \end{cases} \quad (41)$$

The disturbances acting on the end-effector are modeled as bounded perturbations along the x , y , and z -directions, defined as follows:

$$\begin{cases} w_x = 0.6 \sin(5t) \\ w_y = 0.4 \sin(5t) \\ w_z = 0.2 \sin(5t) \end{cases} \quad (42)$$

For the nominal system planning solver, based on the fully constrained cooperative task design in Section 4.1, the parameters setting is shown as following:

Parameters	Simulation setting
predictive datasets	$\{N_p = 5, N_c = 1\}$; $N_{track}^i = I_{12}$; $N_{cop}^i = 0.5I_{12}$; $N_{obj}^i = 0.5I_{12}$
initial state	$[-1.05, 0.5, 0, 0.5, -0.5, -0.5, 0.5, -1.05, -0.5, 0, 0.5, -0.5, -0.5, 0.5]$
tracking term weight matrix	$Q_a = I_{12}$
control variable weight matrix	$R_a = \text{diag}(1, 10^5, 1, 10^5, 1, 10^3, 10^3, 10^3, 1, 10^3, 10^3, 10^3)$
position constraint (upper limit)	$[10, 10, 10, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi, \pi]$
position constraint (lower limit)	$[-10, -10, -10, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi, -\pi]$
velocity constraint (upper limit)	$[1, 1, 1, \pi/6, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36, \pi/36]$
velocity constraint (lower limit)	$[-1, -1, -1, -\pi/6, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36, -\pi/36]$
robust tracking term weight matrix (auxiliary solver)	$Q_{anci} = 100I_{12}$
manipulator-vehicle coordination term (motion priority tube)	$W_{tube} = \text{diag}(1, 1, 1, 10^2, 10^2, 10^2, 10^2, 10^2, 10^2, 10^2, 10^2, 10^2)$

As shown in Fig. 16(a), the traditional MPC-based motion planning method fails to achieve stable capture tasks due to its inability to real-time account for the state error between the nominal model and the actual disturbed model, resulting in oscillatory trajectories around the target. In contrast, as shown in Fig. 16(b), through the FBTMPC method can track the target trajectory, the end-effector's path exhibits redundant movements due to the lack of coordinated compensation design between the vessel and manipulator. This unstable motion pattern not only compromises the successful completion of the capture task but also increases unnecessary energy consumption during operation.

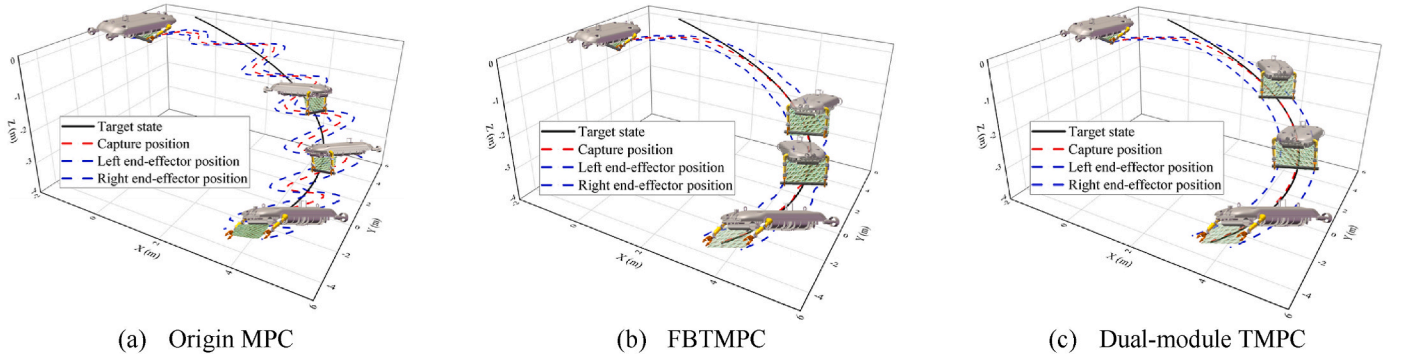


Fig. 16. End-effectors capture trajectory.

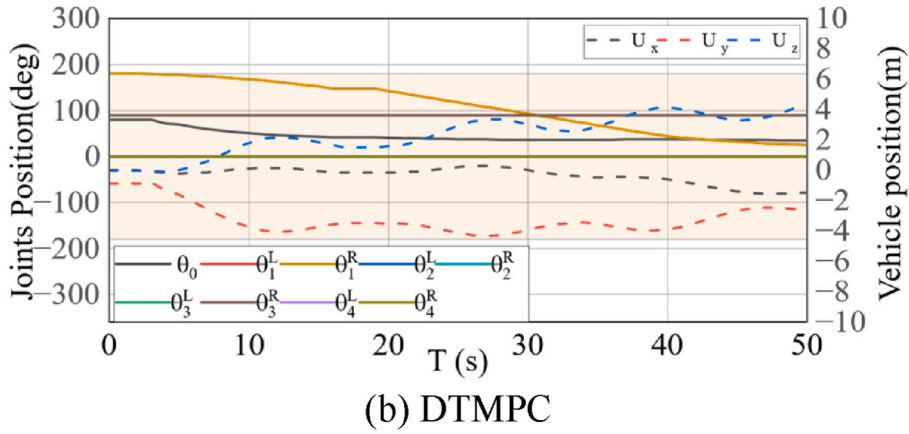
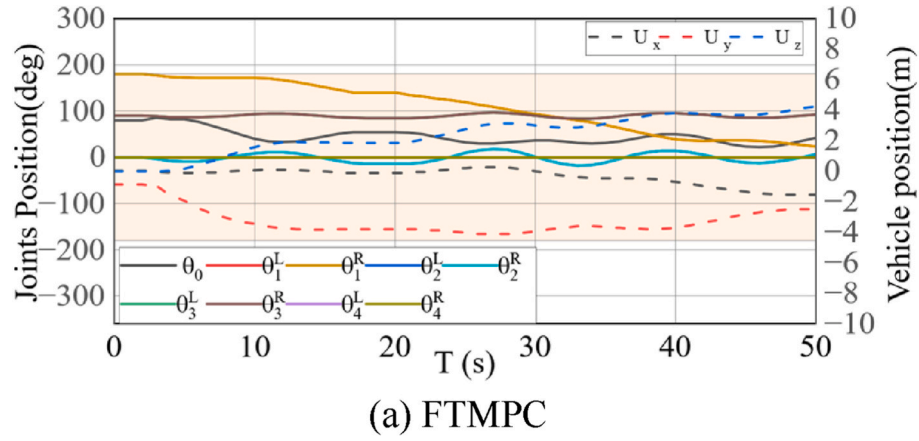


Fig. 17. System joints motion.

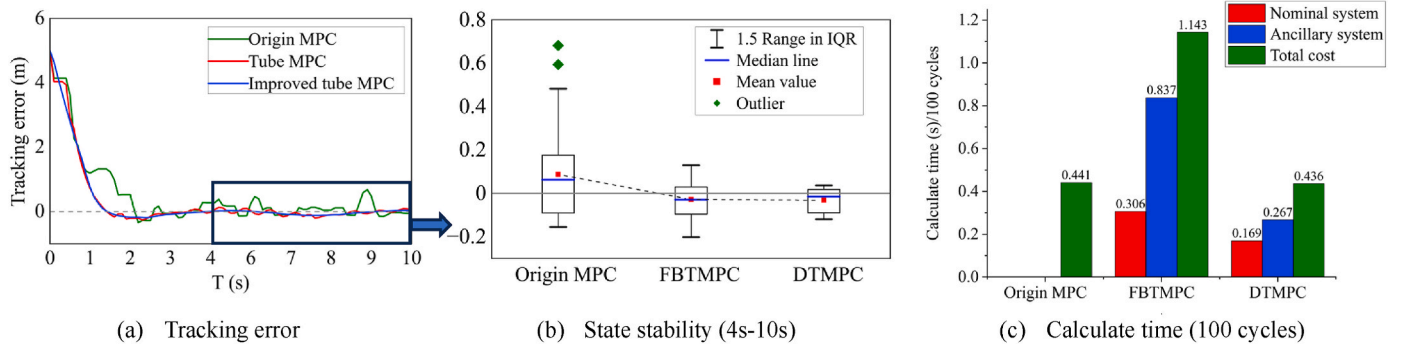


Fig. 18. Planning effect quantitative comparison.

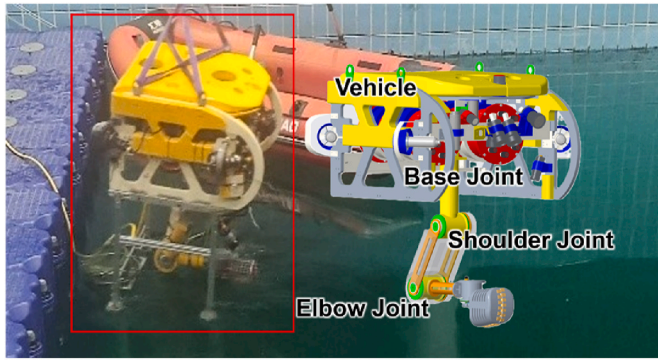


Fig. 19. The structure of used UVMS.

As shown in Fig. 16(c), the proposed method employs coordinated control design between the vessel and manipulator to ensure the planned system state follows the target precisely while effectively compensating for errors caused by external disturbances and internal modeling uncertainties. The coordination mechanism of the DTMP method ensures smooth motion of the robot within the nominal system's planned operational range, significantly reducing redundant movements in the end-effector's trajectory. This enables successful completion of dynamic capture tasks while achieving remarkable improvements in motion accuracy and energy efficiency.

In the simulated disturbance environment, the joint-space planned positions of the system are illustrated in Fig. 17. The auxiliary solver based on state-feedback control, as shown in Fig. 17(a), compensates for the state error between the nominal system and the disturbed system without coordinating the vessel's degrees of freedom with the dual arms' joints relative to the motion priority tube. This lack of coordinated constraint enforcement results in unordered joint adjustments, destabilizing the robot's trajectory. In contrast, as shown in Fig. 17(b), the proposed DTMP method with manipulator-vehicle coordination design demonstrates stable joint-space planning outcomes. By integrating the auxiliary solver design from Section 4.2, the robot achieves coordinated motion between the vessel and manipulator during underwater disturbances, enabling precise error compensation while suppressing redundant movements in the mechanical arms' joints. This ensures successful completion of dynamic capture tasks through optimized motion planning in aquatic environments.

As shown in Fig. 18, a quantitative comparison with the origin MPC and the FBTMP method is conducted to evaluate the effectiveness of the proposed DTMP framework. As shown in Fig. 18(a), the tracking performance over time clearly indicates that the DTMP achieves the most stable and accurate tracking behavior. Once the robot following the moving target, the tracking error of DTMP remains consistently close to zero, with only minor fluctuations.

To highlight the stability of each method under disturbance, we extracted the tracking error data from 4 s to 10 s and plotted the results as a boxplot in Fig. 18(b). The results demonstrate that the original MPC suffers significant tracking deviations due to external disturbances. Although FBTMP maintains similar adherence to the target, it exhibits larger fluctuations compared to DTMP, suggesting limited operational stability. In contrast, the DTMP method yields the narrowest inter-quartile range, confirming its superior stability in tracking.

Furthermore, Fig. 18(c) presents the computational cost over 100 cycles. While FBTMP incurs high computation time due to the need for online state-feedback compensation, DTMP benefits from a model-based auxiliary MPC design. Its use of tube constraints coordinates the redundant degrees of freedom efficiently, reducing the optimization

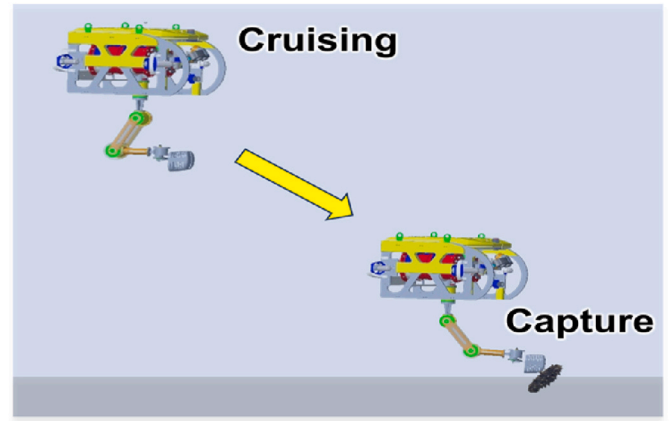


Fig. 20. Pool experimental process illustration.

burden in the auxiliary system. As a result, DTMP achieves lower auxiliary system runtime. Remarkably, the overall computational cost of DTMP is even slightly lower than that of the original MPC, due to its better tracking performance and improved system stability.

In summary, the proposed DTMP method demonstrates excellent planning performance in terms of real-time tracking accuracy, operational stability under disturbances, and computational efficiency. These results validate its practical advantages over baseline MPC-based approaches.

5.3. Preliminary pool experiment

To further validate the effectiveness of the proposed planning method, an experiment was conducted in a pool environment. Since the dual-arm underwater robot is still under development, the experiment utilized a UVMS platform with autonomous underwater grasping capability to test the DTMP motion planning method.

The UVMS platform used in this study consists of an underwater vehicle body and a three-DOF electric manipulator, as shown in Fig. 19. The robot measures 900 mm in length, 750 mm in width, and 1150 mm in height, with a total weight of 85 kg and a maximum operating depth of 100m. The manipulator is integrated into the main frame and comprises three rotational joints. The link lengths of the manipulator are 0.448m, 0.3m, and 0.3m for the base, shoulder, and elbow joints, respectively. A gripper is mounted at the end-effector for grasping operations.

The objective of this pool experiment is:

- (1) Evaluating whether the robot can reach a target object and perform a coordinated motion between the vehicle and the manipulator, following the ideal task process illustrated in Fig. 4 (a): the vehicle first approaches the target, followed by manipulator-based fine adjustment for grasping.
- (2) Examining whether the robot's actual execution trajectory in the underwater environment significantly deviates from the planned trajectory, to assess whether the auxiliary MPC solver effectively handles real-world disturbances.

As shown in Fig. 20, the robot was commanded to maintain a fixed height of 1.8 m and cruise the pool. Upon detecting the target, it descended for grasping. The end-effector's initial position in the world frame was $(-3.4, -7.9, 1.0)$, and the target was located at $(-2.75, -8.5, 0)$. The cost weight for vehicle motion was set to 10, and for manipulator motion to 1.

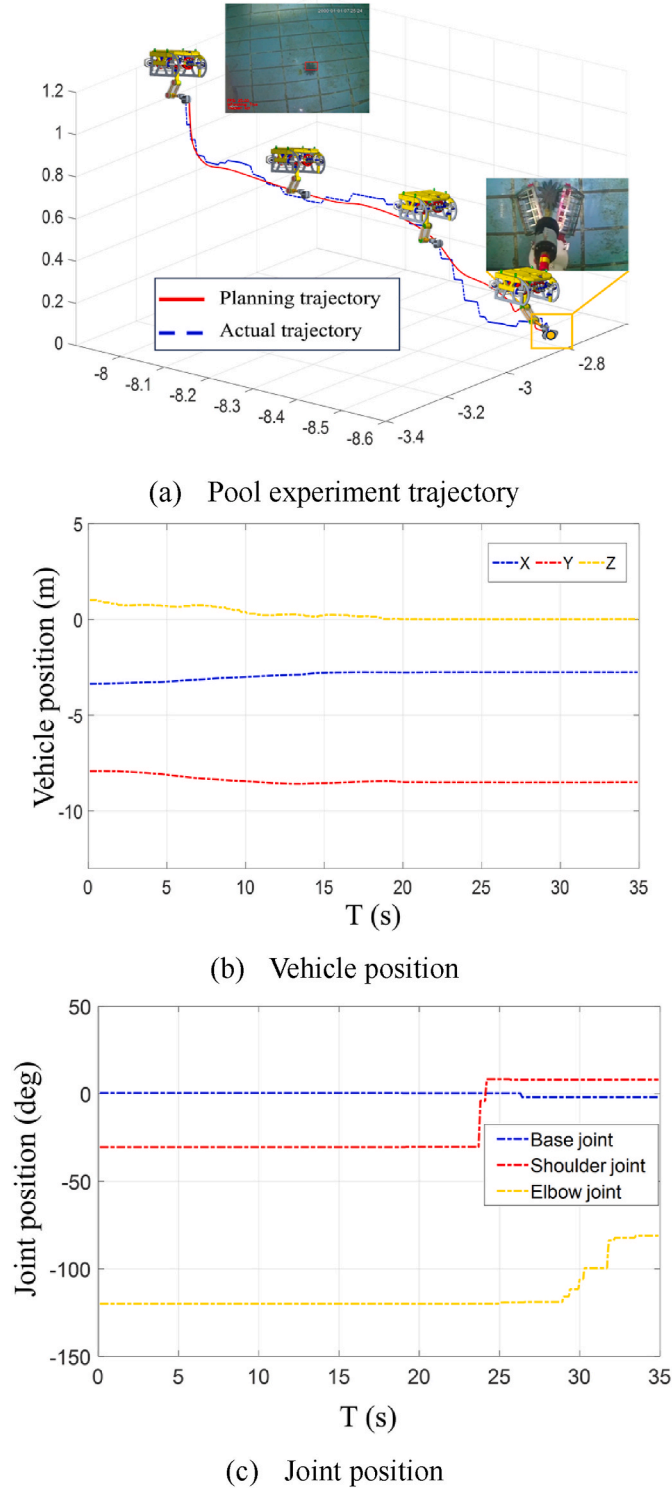


Fig. 21. Pool experiment data.

The planning and actual trajectories are presented in Fig. 21(a). At the early operation stage, the planned trajectory shows a fast descent due to the large initial positional error. During the approach phase, the motion remains smooth due to the optimization term in formula (27), and as the robot nears the target, the manipulator begins to act and quickly aligns with the object. This indicates that the proposed method remains effective even under different model conditions and achieves

the intended manipulator-vehicle coordinated planning objectives.

In the actual trajectory, disturbances such as modeling inaccuracies, tether-induced forces, and internal dynamic coupling effects do influence the system. Nevertheless, the auxiliary MPC compensator ensures that the robot remains close to the ideal trajectory throughout, demonstrating the robustness of the proposed planning method in a real underwater environment. Moreover, as Fig. 21 shown, onboard system logs confirm that the robot first uses vehicle movement to approach the target, followed by manipulator movement to complete the task—verifying the intended manipulator-vehicle coordination process.

These results demonstrate that the proposed planning framework not only performs well in simulation but also retains robustness and adaptability under real-world underwater conditions. This lays a strong foundation for future deployment on more complex dual-arm underwater platforms.

6. Conclusion

This paper addresses the motion planning problem for underwater dual-arm robots in cooperative tasks. By systematically analyzing different cooperative motion modes, we investigate the coordination between the vehicle and dual arms, as well as inter-arm cooperation. Combining robust planning concepts, we achieve successful underwater cooperative operations for dual-arm robots. For dynamic cooperative tasks in disturbed environments, we propose a robust motion planning method based on Dual-Module Tube MPC. Simulation and pool experiment results demonstrate the effectiveness of the proposed approach.

However, this study requires further exploration regarding target information acquisition and processing in planning tasks. Integrating robot perception technologies for real-time planning could significantly reduce positioning errors, which holds critical importance for underwater operations. Therefore, future research will focus on combining perception systems with motion planning to enhance localization accuracy. Constructing comprehensive dynamic task models for complex underwater environments is also essential for improving motion planning capabilities. Subsequent investigations will delve deeper into dynamic information modeling in aquatic scenarios.

CRedit authorship contribution statement

Yize Sun: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Hai Huang:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Hongde Qin:** Resources, Funding acquisition. **Xianglong Wen:** Software, Formal analysis. **Hao Zhou:** Supervision, Methodology.

Declaration

The authors declare that there is no conflict of interests regarding the publication of this article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX

The underwater dual-arm robot studied in this paper is a humanoid UVMS equipped with two underwater manipulators. The two four-joint homogeneous manipulators are symmetrically distributed on both sides of the robot's hull. A kinematic model for the dual manipulators is established using the modified D-H parameter method.

Table 4
Integrated kinematic model parameters

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	d_1	0
2	90°	0	d_2	90°
3	90°	0	d_3	90°
4	90°	0	0	θ_0
5(9)	-90°	0	$l_0 (-l_0)$	$\theta_1 (\theta_{1R})$
6(10)	90°	l_1	0	$\theta_2 (\theta_{2R})$
7(11)	0	l_2	0	$\theta_3 (\theta_{3R})$
8(12)	90°	0	l_3	$\theta_4 (\theta_{4R})$

where link length $l_0 = 0.5$ m, $l_1 = 0.25$ m, $l_2 = 0.5$ m, $l_3 = 0.3$ m.

Following the rules of the modified D-H coordinate system, the coordinate system at the base is defined as the base frame. The shoulder joint, upper arm joint, forearm joint, and wrist joint are connected sequentially to form the dual-arm parameter coordinate system as shown in Fig. 22:

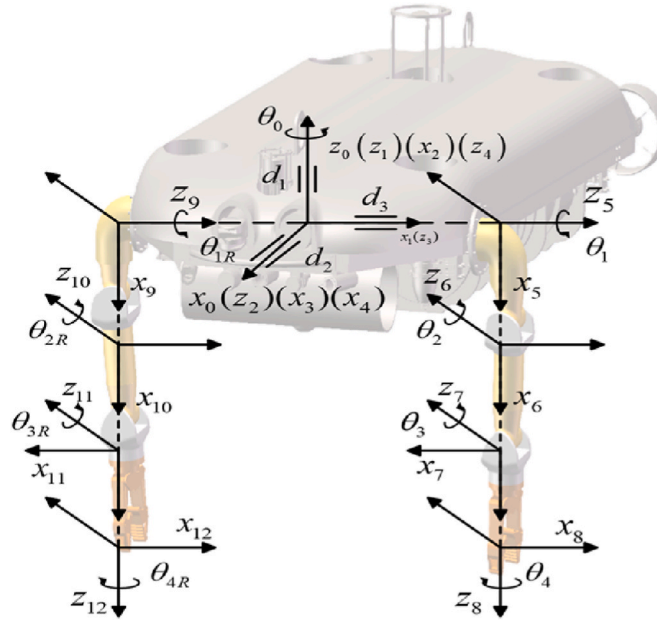


Fig. 22. The structure of dual-arm I-AUV

The relationship between two adjacent coordinate systems can be represented by the relationship between adjacent links in the modified D-H parameter method, which is expressed as:

$${}^{i-1}T_i = Rot(X, \alpha_{i-1}) Trans(a_{i-1}, 0, 0) Rot(Z, \theta_i) Trans(0, 0, d_i) \quad (43)$$

$$= \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i) & 0 & a_{i-1} \\ \sin(\theta_i)\cos(\alpha_{i-1}) & \cos(\theta_i)\cos(\alpha_{i-1}) & -\sin(\alpha_{i-1}) & -\sin(\alpha_{i-1})d_i \\ \sin(\theta_i)\sin(\alpha_{i-1}) & \cos(\theta_i)\sin(\alpha_{i-1}) & \cos(\alpha_{i-1}) & \cos(\alpha_{i-1})d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Based on the modified D-H method, the integrated kinematic model of the dual-arm I-AUV is obtained:

$${}^0T_n^L = {}^0T_1^L T_2^L \dots T_{n-1}^L T_n^L = \begin{bmatrix} r_{11}^L & r_{12}^L & r_{13}^L & p_x^L \\ r_{21}^L & r_{22}^L & r_{23}^L & p_y^L \\ r_{31}^L & r_{32}^L & r_{33}^L & p_z^L \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (44)$$

$${}^0T_n^R = {}^0T_1^R T_2^R \dots T_{n-1}^R T_n^R = \begin{bmatrix} r_{11}^R & r_{12}^R & r_{13}^R & p_x^R \\ r_{21}^R & r_{22}^R & r_{23}^R & p_y^R \\ r_{31}^R & r_{32}^R & r_{33}^R & p_z^R \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (45)$$

where $p_x^L, p_y^L, p_z^L, p_x^R, p_y^R, p_z^R$ represent the position of the end-effectors; the 3×3 matrix of $r_{ij}^{R/L}$ represent the posture of the end-effectors.

Considering the notable advantage of quaternions in effectively avoiding singularity issues when handling 3D spatial rotations, this study also attempted to use the quaternion-based method in simulations to describe the end-effector pose of the underwater dual-arm robot's manipulator, thereby ensuring the generality of motion planning during the independent execution of diverse multi-task objectives by the two arms.

The analytical expression for the conversion relationship between quaternions and the elements of the rotation matrix (Carmeli and Malin, 2000) is as follows:

$$\begin{cases} |q_1| = \frac{1}{2} \sqrt{1 + r_{11} + r_{22} + r_{33}} \\ |q_2| = \frac{1}{2} \sqrt{1 + r_{11} - r_{22} - r_{33}} \\ |q_3| = \frac{1}{2} \sqrt{1 - r_{11} + r_{22} - r_{33}} \\ |q_4| = \frac{1}{2} \sqrt{1 - r_{11} - r_{22} + r_{33}} \\ \text{sign}(q_2) = \text{sign}(q_1) [\text{sign}(r_{32} - r_{23})] \\ \text{sign}(q_3) = \text{sign}(q_1) [\text{sign}(r_{13} - r_{31})] \\ \text{sign}(q_4) = \text{sign}(q_1) [\text{sign}(r_{21} - r_{12})] \end{cases} \quad (46)$$

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